

CO2020 : Computer-Aided Numerical Methods II

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PDEs Homework 1

Due Date: 9 March 2026, Monday, 11 pm

Note: In this document, subscript denotes a differentiation, e.g. u_x denotes $\partial u / \partial x$. Variables such as x, y, z, t are generally independent variables. The unknown u is a function of the independent variables as per context. Any other symbol (e.g. c, ν) is a constant unless otherwise stated.

1. State the order of the following PDEs and whether they are homogeneous/heterogeneous and linear/non-linear.

(a) $u_{xx} + u_{yy} = 0$

(b) $u x u_x + u^2 x y u_{xy} = 0$

(c) $(u_t)^2 + f(x, y) u = g(x, y)$

(d) $u_t + u u_x = \nu u_{xx}$

(e) $u_t + c u_x = 0$

2. Identify whether the 2nd order PDEs given below are elliptic, parabolic or hyperbolic. If an equation is parabolic or hyperbolic, state the number of real characteristics and find their expressions.

(a) $x^2 u_{xx} - 2xy u_{xy} + y^2 u_{yy} = 0$

(b) $u_{xx} + (1 + y^2)^2 u_{yy} - 2y(1 + y^2) u_y = 0$

(c) $x^2 u_{xx} - 2xy u_{xy} - 3y^2 u_{yy} = u_y - u_x + f(x, y) u$

3. Solve analytically the Laplace equation $T_{xx} + T_{yy} = 0$ for the temperature on a rectangular domain of size $a \times b$. The horizontal boundaries ($y = 0$ and $y = b$) are insulated, and the left ($x = 0$) and right ($x = a$) boundaries are held at T_0 and $(T_0[1 + 0.2(y/b - 0.5)])$ respectively. For $a = 1$ m, $b = 0.5$ m and $T_0 = 300$ K, plot (a) contours in the 2D domain; (b) profiles of T vs y for $x/a = 0.1, 0.3, 0.5, 0.7$ and 0.9 ; (c) profiles of T vs x for $y = 0.1, 0.5$ and 0.9 . Generate one figure each for parts (a), (b) and (c).

4. The 2D steady heat conduction equation,

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \dot{q} = 0,$$

is to be solved using a second-order accurate finite difference method on the domain $[0, Lx] \times [0, Ly]$. The domain is discretized into $nx \times ny$ grid points. The grid points are $x_i = i \times h_x$ with $i = 0, 1, 2, \dots, nx - 1$ and $h_x = Lx/(nx - 1)$ in x and similarly in y . We assume that the diffusivity is $k = 1$ and the source term is $\dot{q} = 0$ everywhere. Impose the same boundary conditions as in Question 3 and set $Lx = 1$ and $Ly = 0.5$.

- (a) Starting from the skeleton code given to you, write a code to solve this problem numerically using the GS-SOR solver.
- (b) Do the calculations for computing the coefficients by hand for a small system, e.g. $nx = 5$ and $ny = 3$. Ensure that your hand calculations agree with the answers computed by the code. You need not report anything for this part. Once you have ensured correctness of the code, you can comment out any unnecessary 'printf' statements so that the output of the code remains clean.
- (c) Run the code for about 5 or 6 grids such as 17×9 , 33×17 , 65×33 , 129×65 etc. For the finest grid that you select, plot the contours of the numerical solution. Also generate line plots of T vs y at $x = 0.9$ and T vs x at $y = 0.4$ with the exact solution from question 3 overlaid with the numerical solutions on different grids. In all, you need to show one contour plot and 2 line plots.
- (d) Calculate the error norm for different grid sizes and hence verify the order of accuracy. This should be reported as a plot of the error norm vs the number of grid points along one direction (or vs the grid spacing).

General instructions:

1. For plotting figures, use Matlab or Python any other postprocessing software of your choice. Ensure that the font size of the legend and labels is large enough to be easily visible. Ensure that the line thickness is appropriately large. Export each image as a png or eps file and include them in a pdf document.

2. Scan your entire handwritten work to pdf, combine it with the pdf containing the figures for Q3 and Q4, along with very brief explanatory comments (one or two sentences) and submit the entire homework as a single pdf on google classroom. Zip all your codes and submit them along with your pdf.